



# It Takes a Village

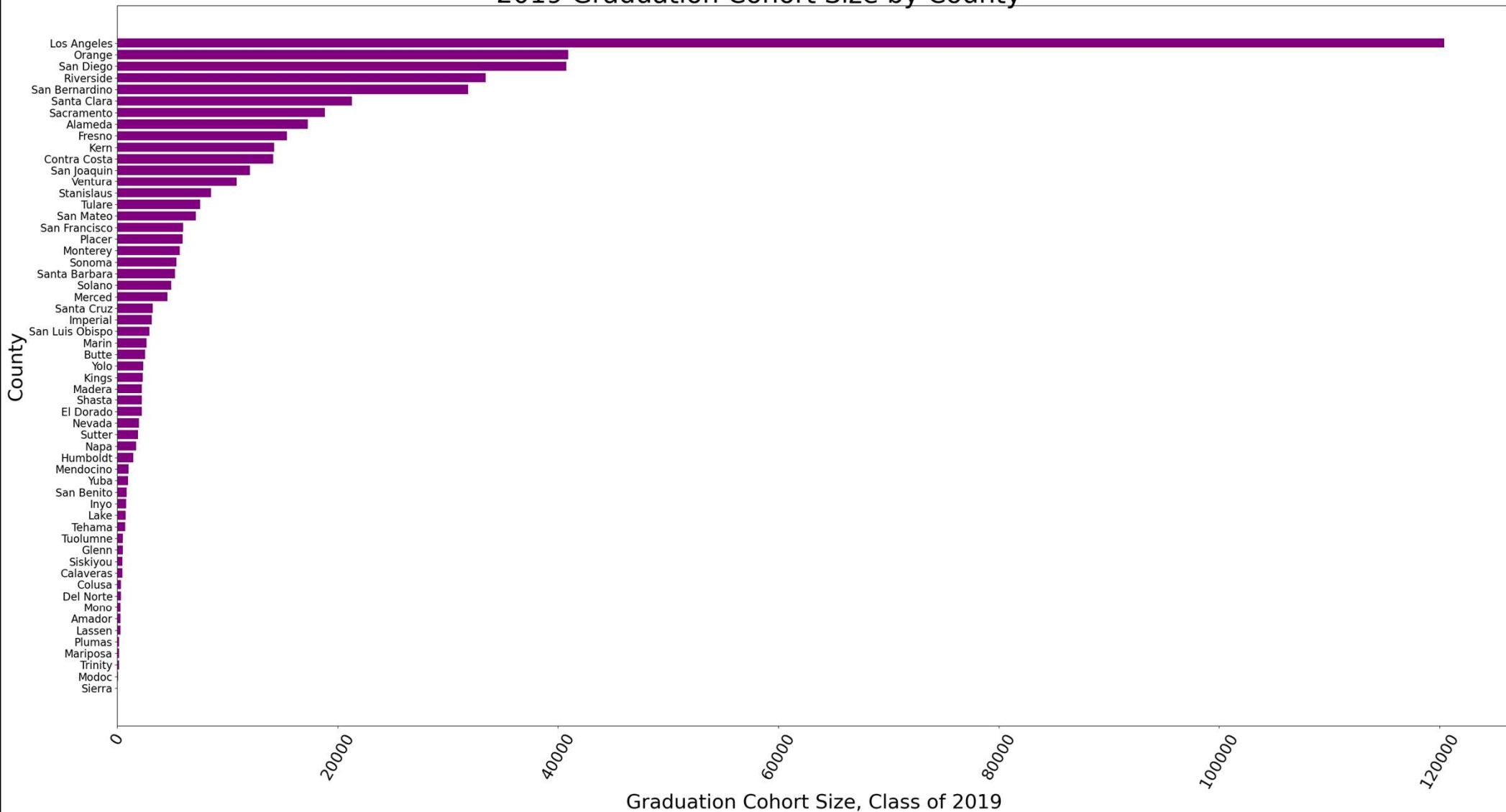
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*A Presentation for the Coalition for Unprecedented Transformative Education for California Teachers and Students, 2024 Meeting*

This is the project on the county-level high school dropout rate in California. The *Coalition for Unprecedented Transformative Education for California Teachers and Students* (which shortens to CUTECATS) is a fake organization we made up as a framing device for why we were giving a non-technical presentation to our classmates and teachers, who all, in reality, had at least as much technical knowledge as we did.

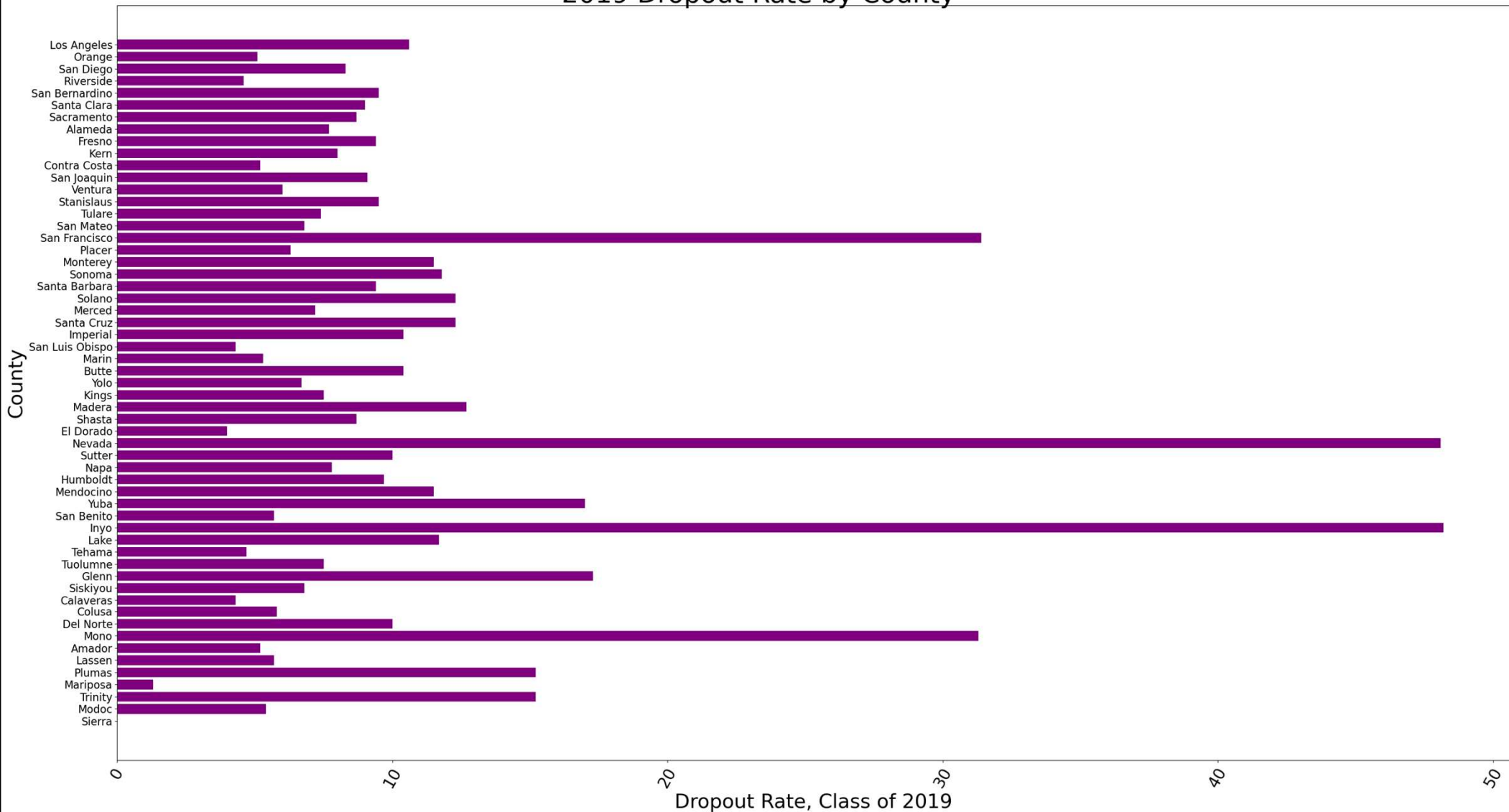
To focus on the visualizations, I have cut out many of the text slides. The full presentation can be found in the PDF in the github repository.

# 2019 Graduation Cohort Size by County



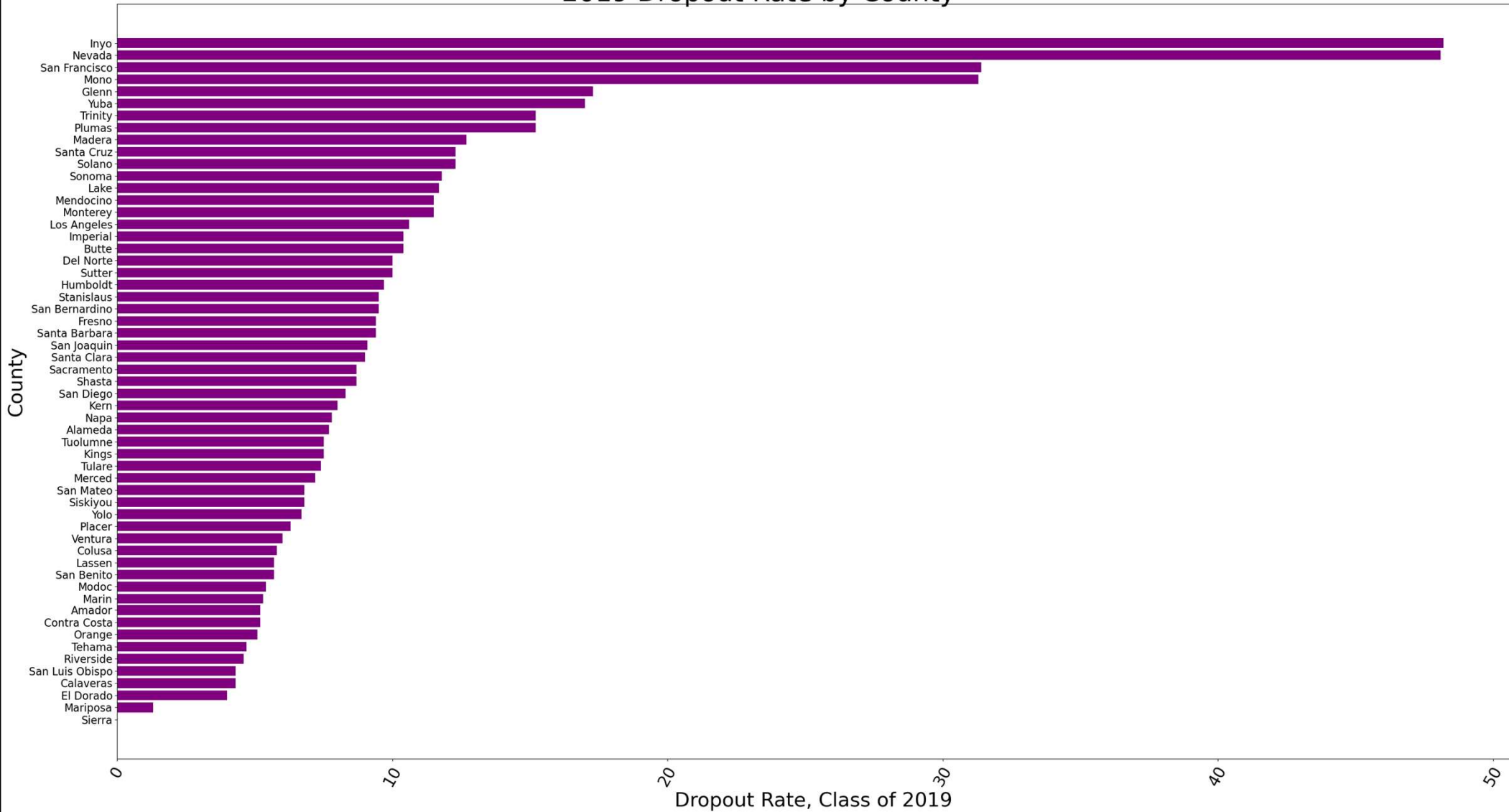
This horizontal bar graph shows the size of the expected graduating class of each county, ordered from largest to smallest to make the disproportionate size of LA county clear. Ordinarily I would format the graph to make the names of the counties easier to read, but as I explained in the presentation, the enormous range of entries made that very difficult. However, it doesn't really matter which bar is which, at least not for a general overview of the project; the purpose of this graph was to show how huge LA County was, how small other counties were, and set up what a problem that was going to be. When discussing this graph, I told people verbally that LA was at the top with over 100,000 students in its expected graduating class, followed by Orange and San Diego counties, whereas Trinity, Modoc, and Sierra counties were at the bottom, with Sierra county having only 23 students in its expected graduating class.

## 2019 Dropout Rate by County

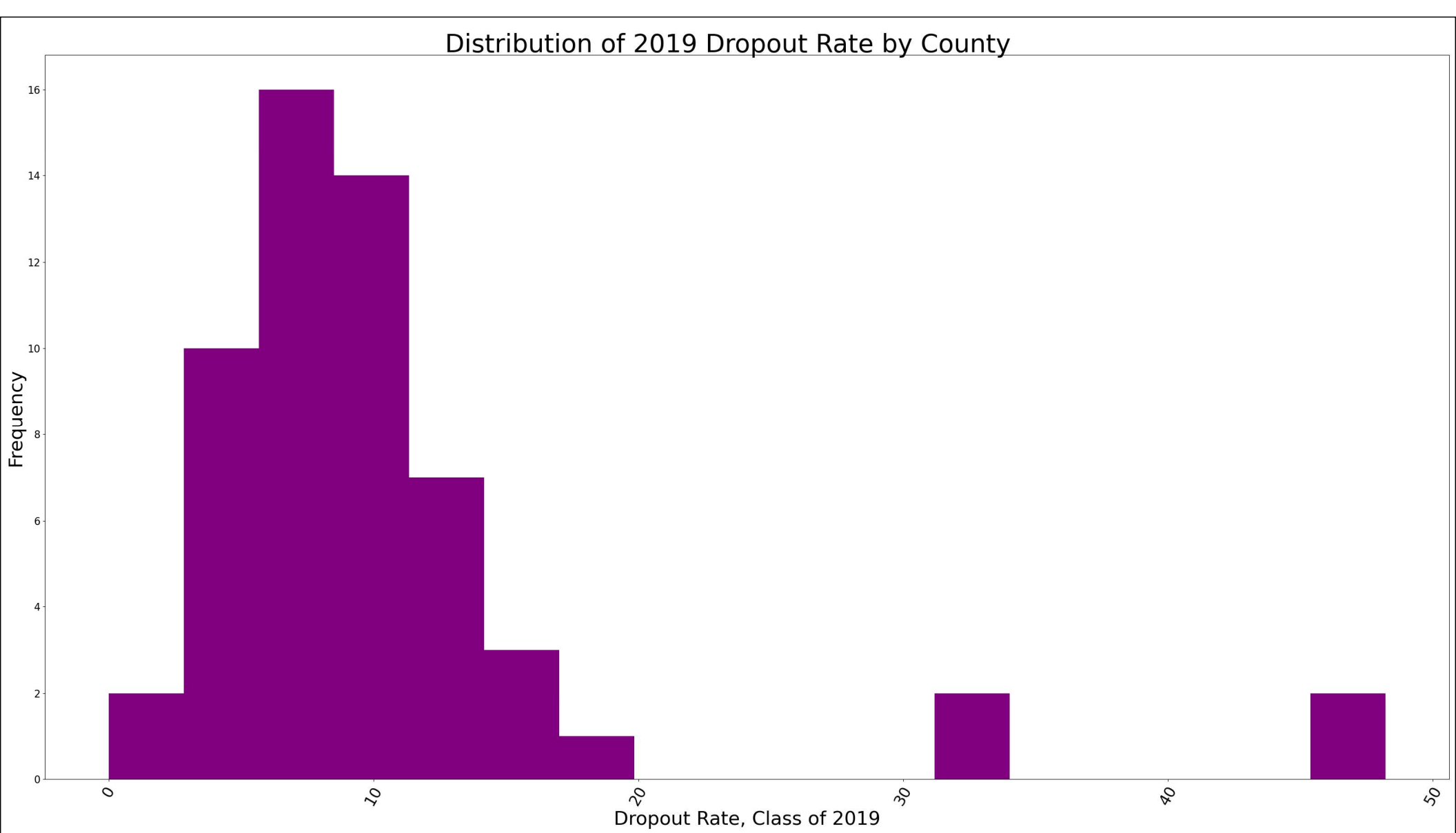


This graph retains the order of the counties from the previous slide, but shows their dropout rates rather than their graduating class size. I believe it is helpful to offer people a 1-1 comparison – in this case, to see that dropout rate is not simply a function of population – but it’s hard to read, so I moved through this slide quickly.

# 2019 Dropout Rate by County



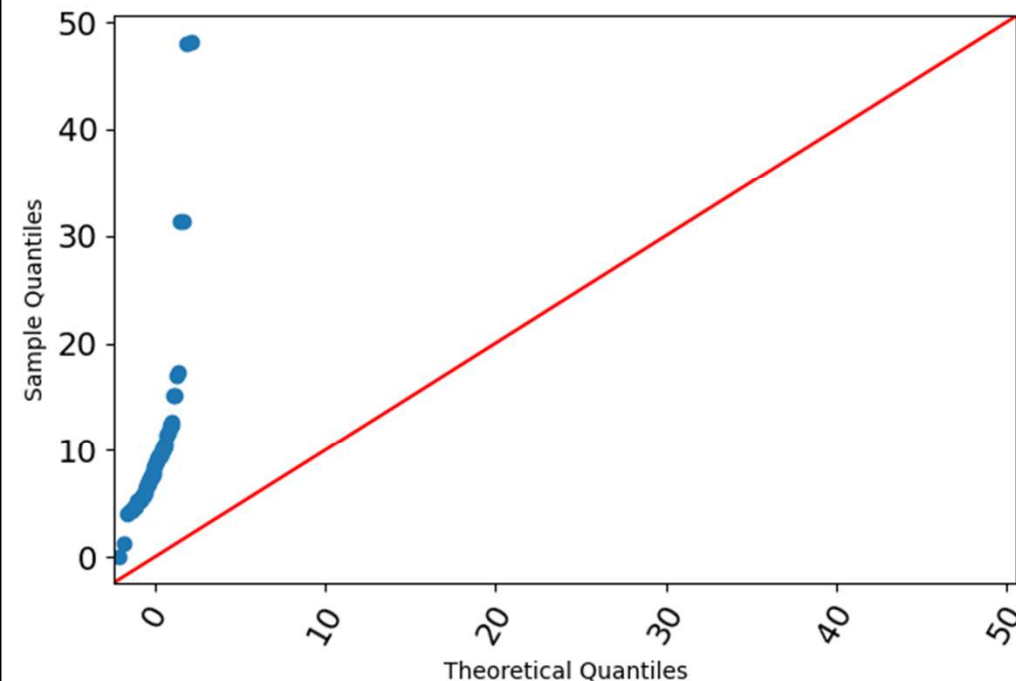
This is the same data from the previous slide, but reordered from highest to lowest rate, to make it easier to read. This graph shows that Inyo and Nevada counties have high dropout rates, nearly 50%, while all 23 seniors in tiny Sierra County graduated. Giant LA County is in the middle.



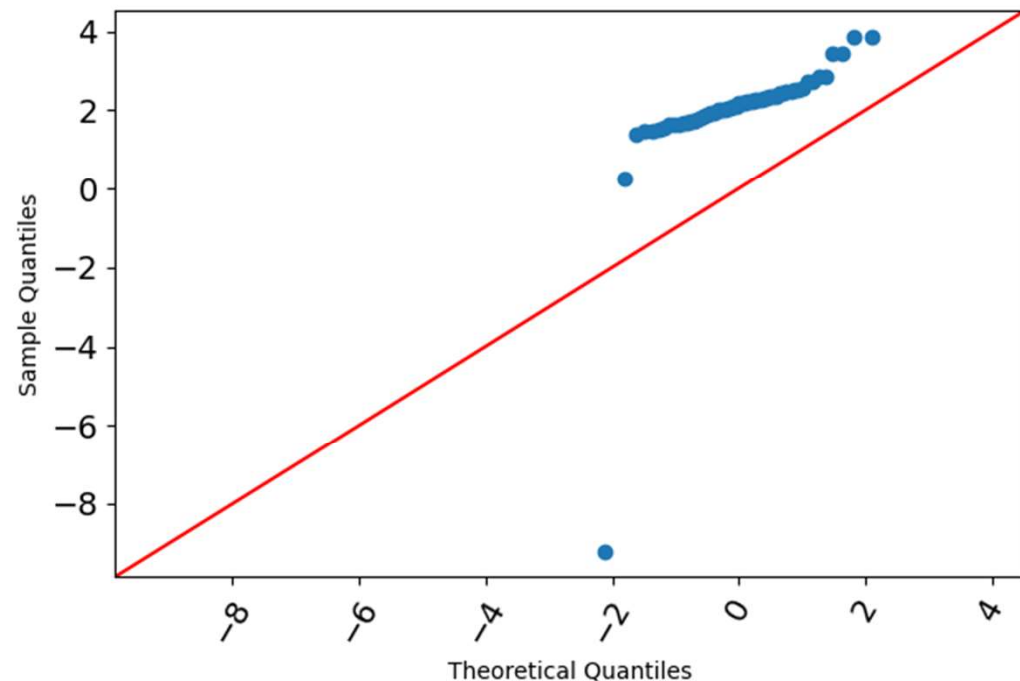
This graph was cut from the original presentation since its content was well covered elsewhere, but it is in the 03\_images folder. This is a histogram of the per-county dropout rates. Whereas the bar graphs of the past few slides show the rate for each county, this histogram shows the frequency of each rate in the dataset. This is not the worst non-normal distribution in this dataset, but the two outliers are an omen of what is to come. Because the dropout rate was the target variable, I attempted to normalize the distribution through transformations (see next slide).

# Distribution Woes

QQ-Plot of 2019 Dropout Rate by County

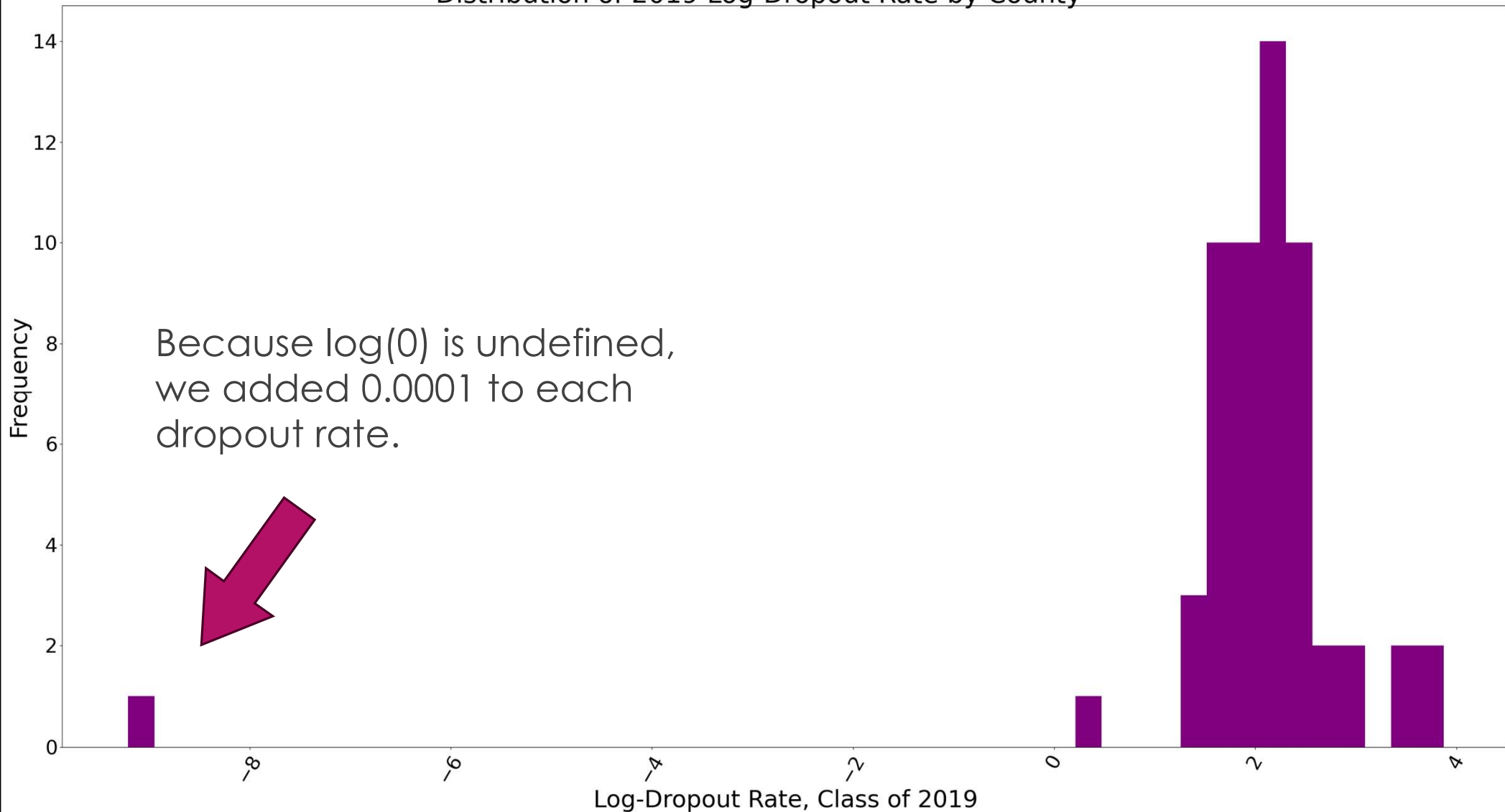


QQ-Plot of 2019 Log Dropout Rate by County



This slide shows QQ-plots of the dropout rate, a way of testing the normality of a distribution. In a normal, amenable-to-modeling distribution, the blue dots would sit neatly on the red line. The plot on the left, of the raw rates, shows that not only do they not sit on the line, but they actually resemble an entirely different distribution (cubic). The plot on the right shows the log of the dropout rates, the most promising method I tested. The blue dots still do not sit on the red line, but they are no longer taking the shape of a different distribution, are more or less in a straight line (albeit in the wrong place), and are closer to the red line. For these reasons, we conducted our analyses using the log of the dropout rates, rather than the raw rates. Our model's predictions can be easily transformed back to percentages by raising them to the  $e$  power.

Distribution of 2019 Log-Dropout Rate by County

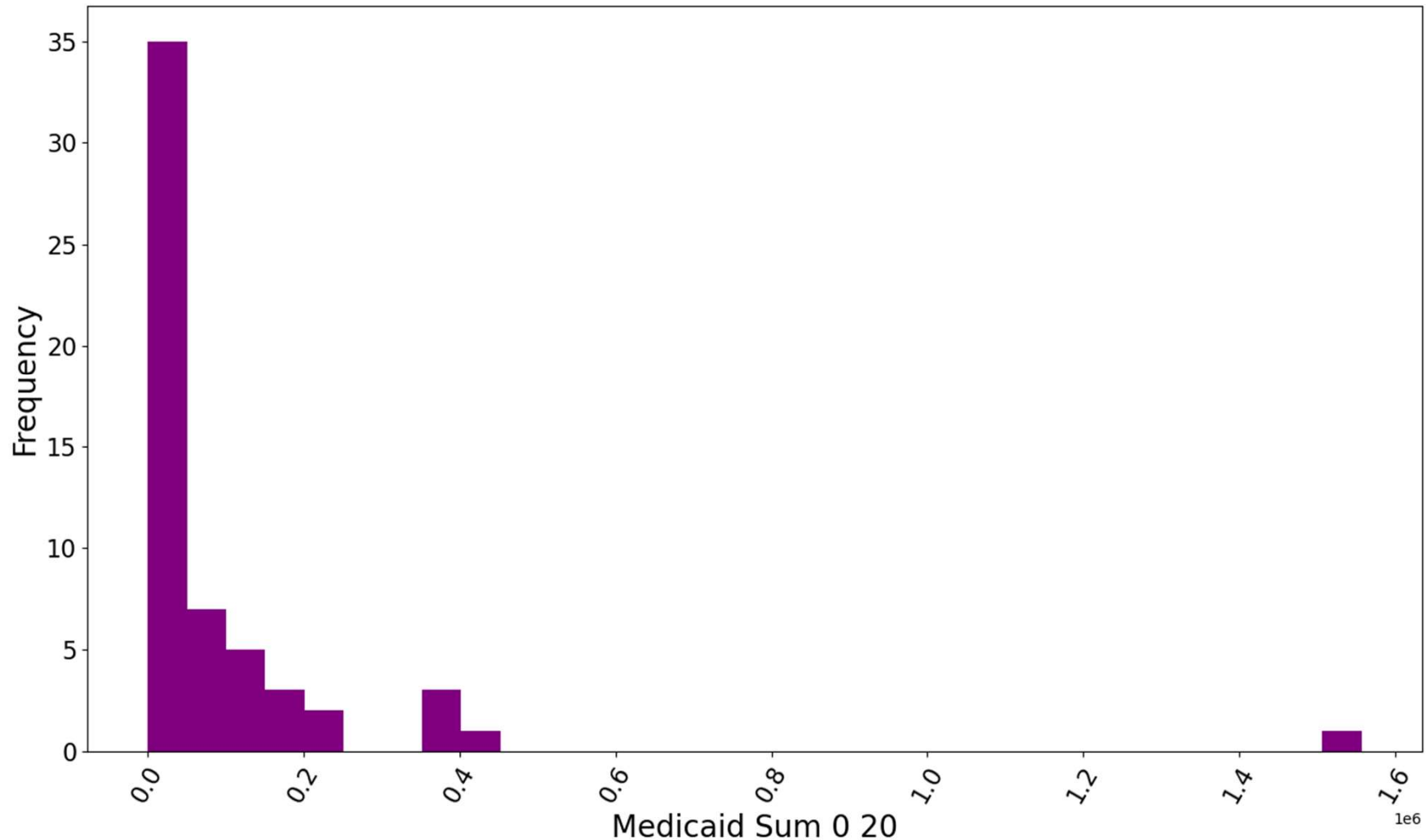


This is a histogram of the log-dropout rates, showing how frequently each value appears in the dataset. As you can see, it is still not a normal distribution, but it is closer. However, in solving the problem of LA County being an outlier, we incurred a problem of Sierra County being an outlier. Thankfully, Sierra County is an outlier on a much smaller scale than LA was.

Before continuing, I'd like to say that this is how I typically format my graphs – with labels that are large enough to be read easily, and written in a clear way rather than using the abbreviated “behind the scenes” variable names. Some of the following graphs do not meet all of those goals, because once it was clear that there were fatal flaws in the data, my team and I opted to heavily prioritize our limited time on addressing that. I gave a verbal explanation of each graph during the presentation.

## Distribution of Medicaid Sum 0 20

Based on 57 Observations out of 57

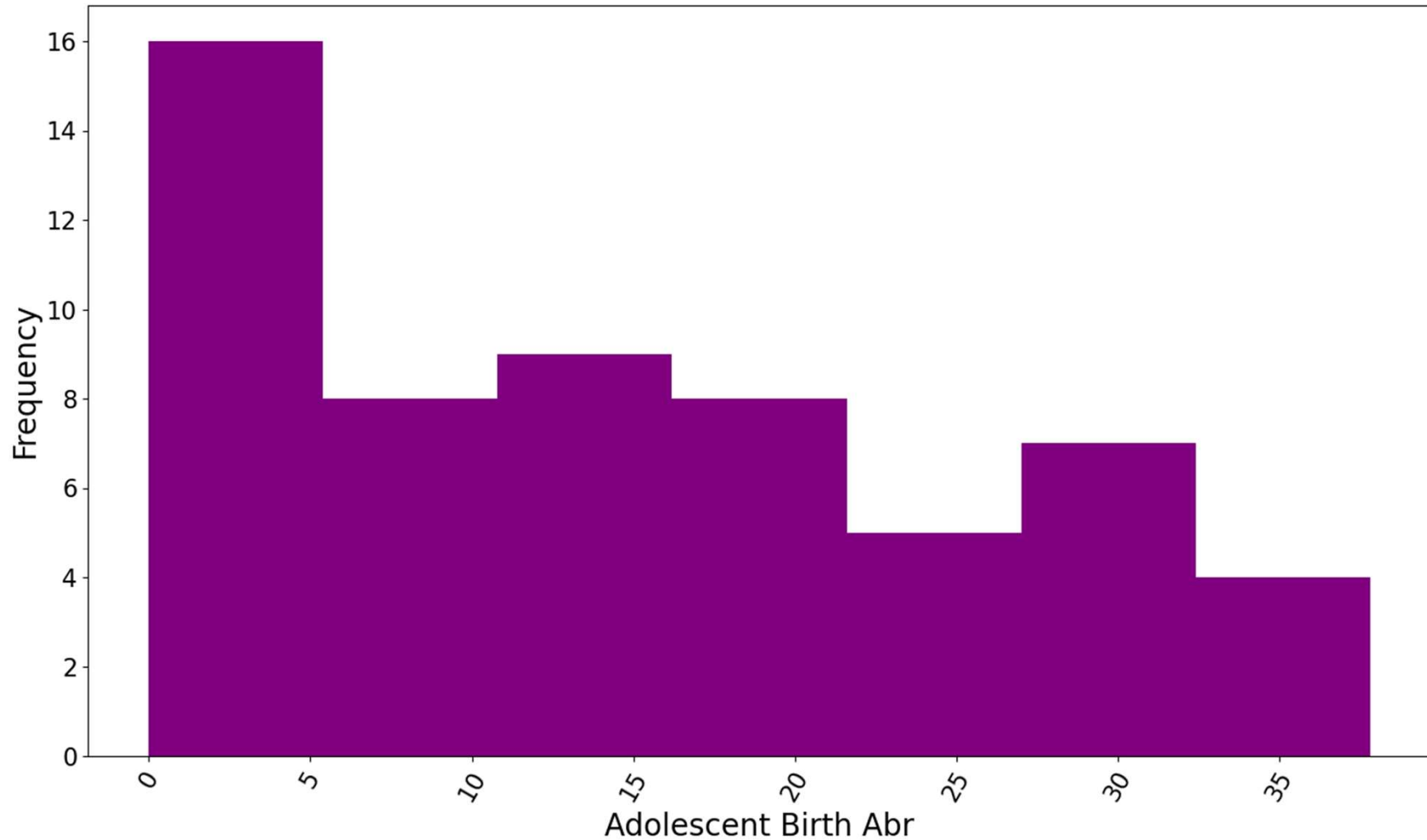


These are the “L not a bell” plots. This one in particular is the log of the number of people under age 21 enrolled in Medicaid per county.

I pulled out some examples to demonstrate, but basically all of our histograms looked like this. Smaller values, seen in the normal-sized counties, produced a huge spike right around 0, then there was a trickle of somewhat larger values, a large blank space, and then LA County at the far end representing one observation of an enormous value for any given variable. LA County is just so big, with so much going on, that it threw off every single variable. It wasn't always this extreme...

## Distribution of Adolescent Birth Abr

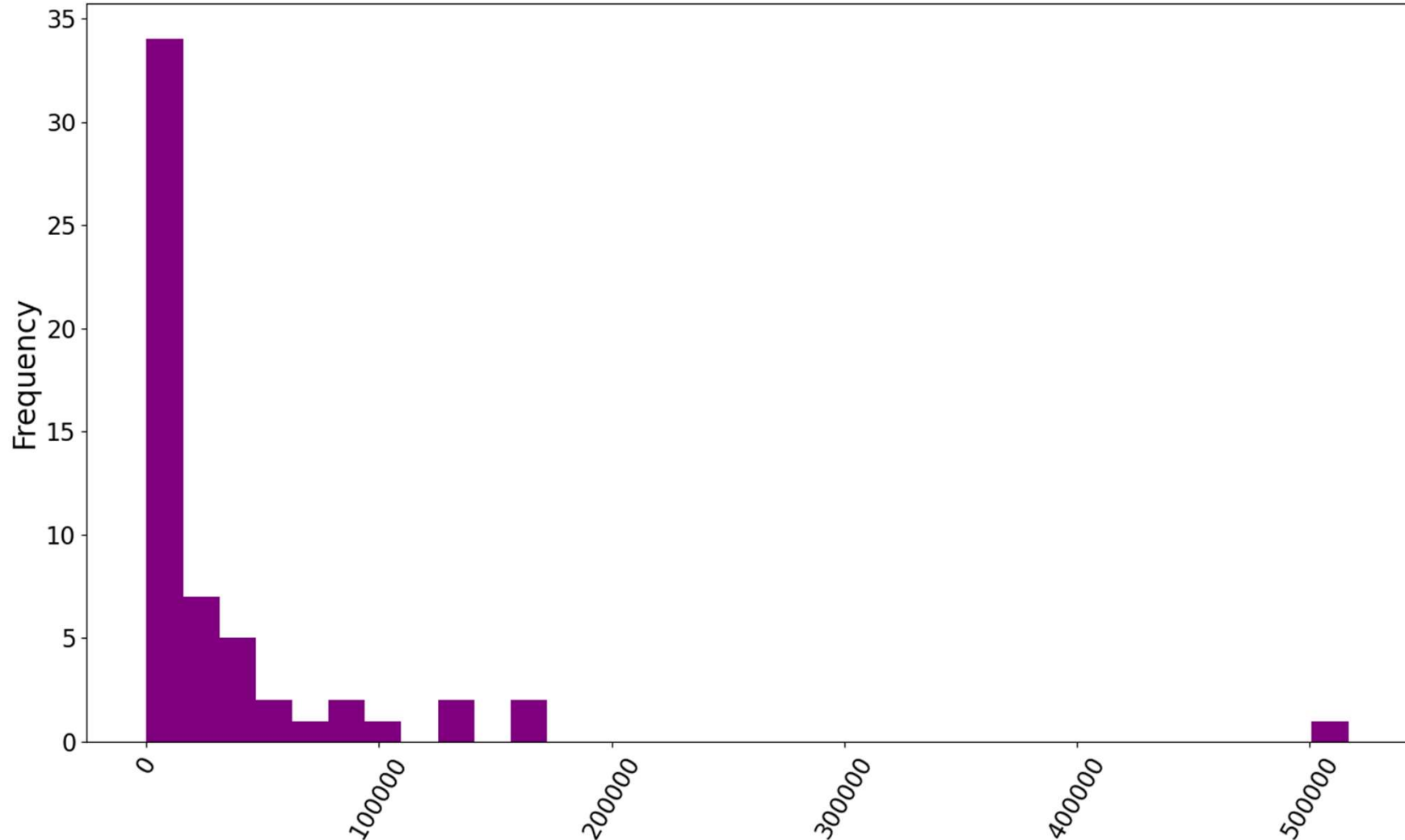
Based on 57 Observations out of 57



This is more reasonable, although not normally distributed by any stretch of the imagination. Most of the histograms were straightforwardly L shaped. This graph shows the distribution of births to people aged 15-19, per capita, per county.

## Distribution of Daycare Slots Child Total Pop

Based on 57 Observations out of 57

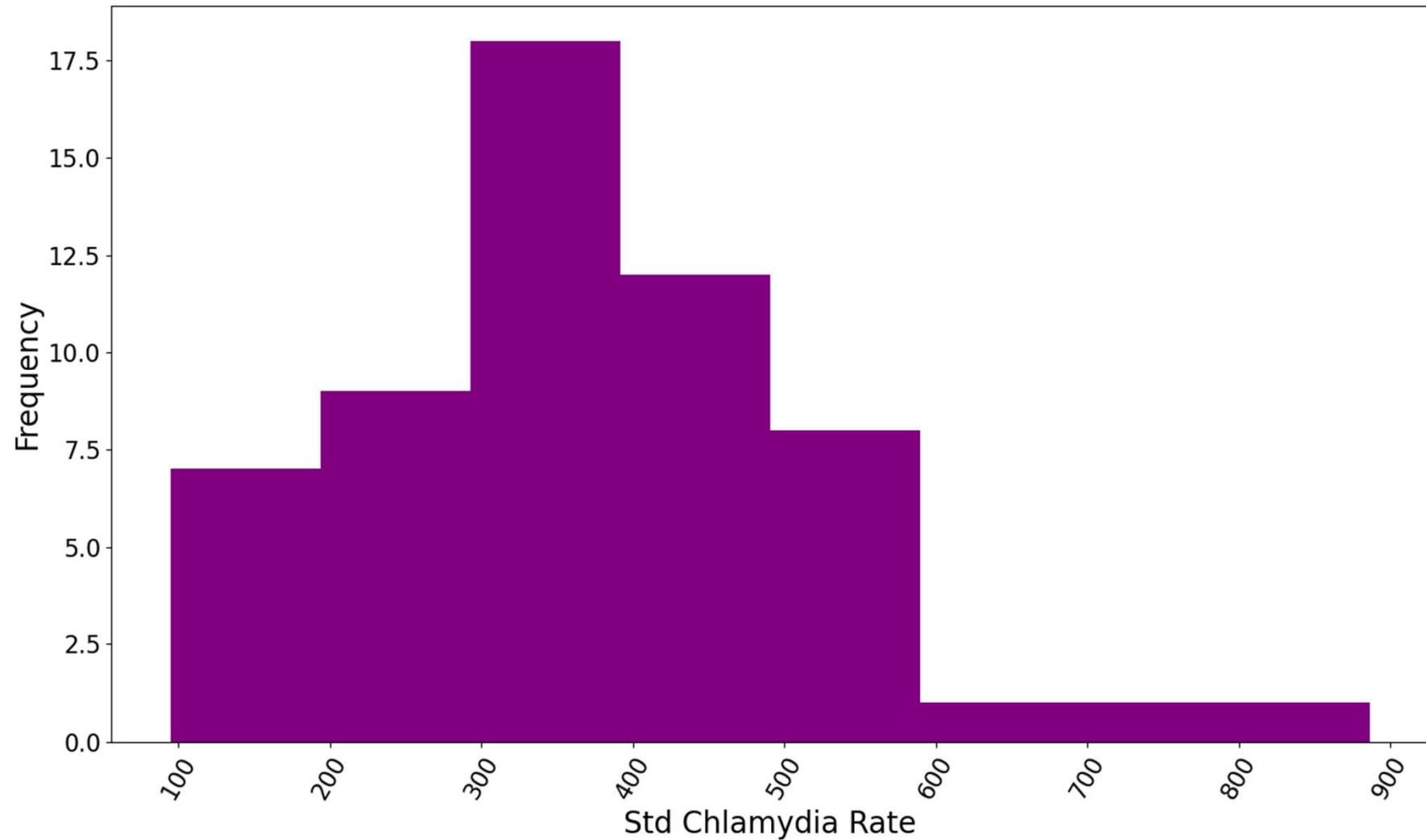


As I stated in the presentation, my personal favorite “L not a bell” graph is this one, because we’re here trying to assess kids dropping out of high school, but based on this graph, most counties have 0 children! Problem solved!

This graph represents the size of the population of children in each county, and this data comes from a larger dataset on the availability of childcare. It is another very good example of small counties being clustered together on the left, and LA County sitting by itself far to the right.

## Distribution of Std Chlamydia Rate

Based on 57 Observations out of 57



This distribution of chlamydia rates per county was the most normally distributed looking variable we had, which may be why it emerged as one of very few statistically significant predictors in our models.

# Results

- ▶ Correlations do not imply causation, but...
- ▶ Availability of ecigarettes seems bad
- ▶ Moreso in combination with other factors than alone

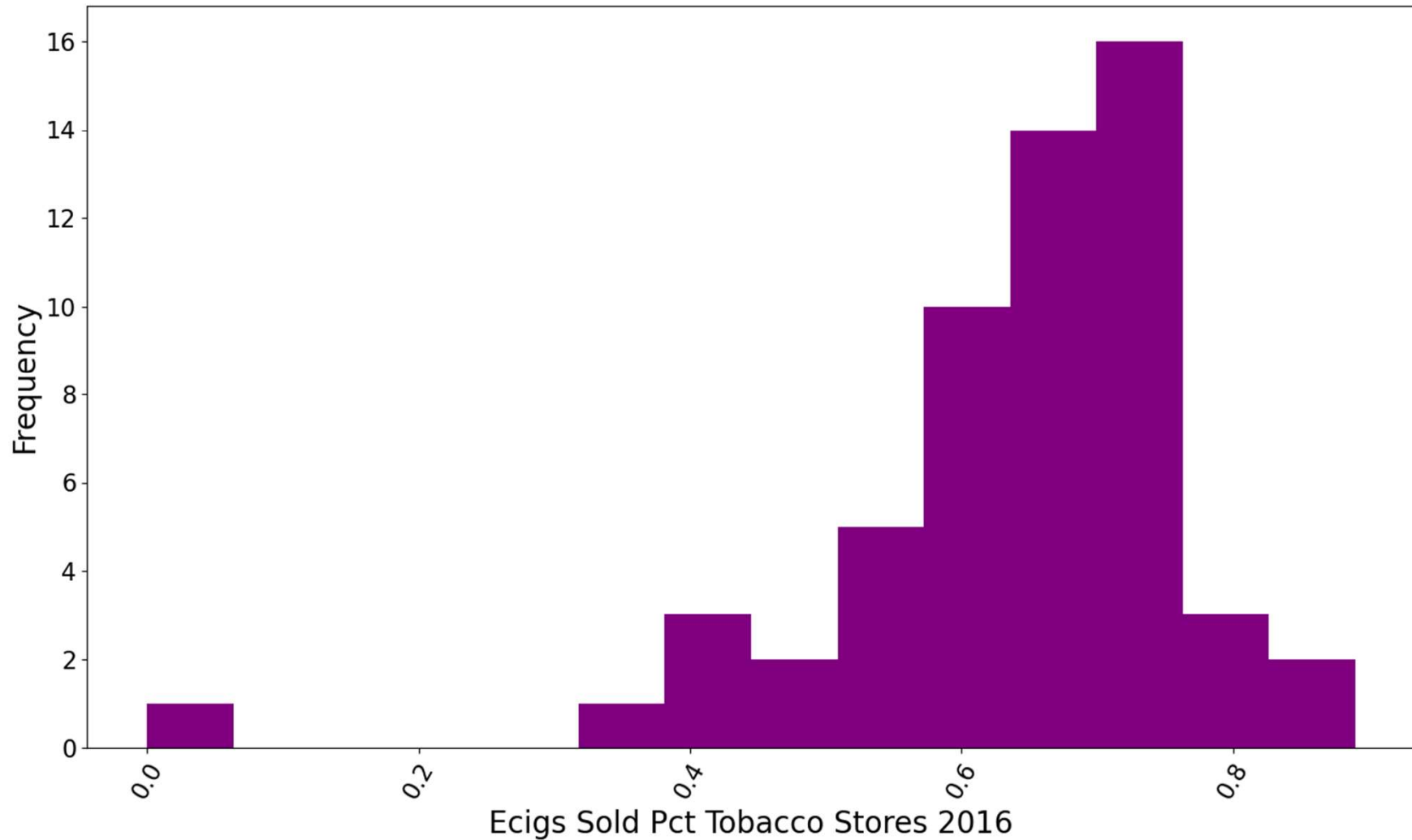


Photo credit: [Vaping360](#) , CC BY 2.0

Due to the challenges in our data, we were not able to identify many useful predictors of the high school dropout rate. One that emerge was the availability of ecigarettes, particularly when used in an interaction term with other variables. More availability predicted higher dropout rates.

# Distribution of Ecigs Sold Pct Tobacco Stores 2016

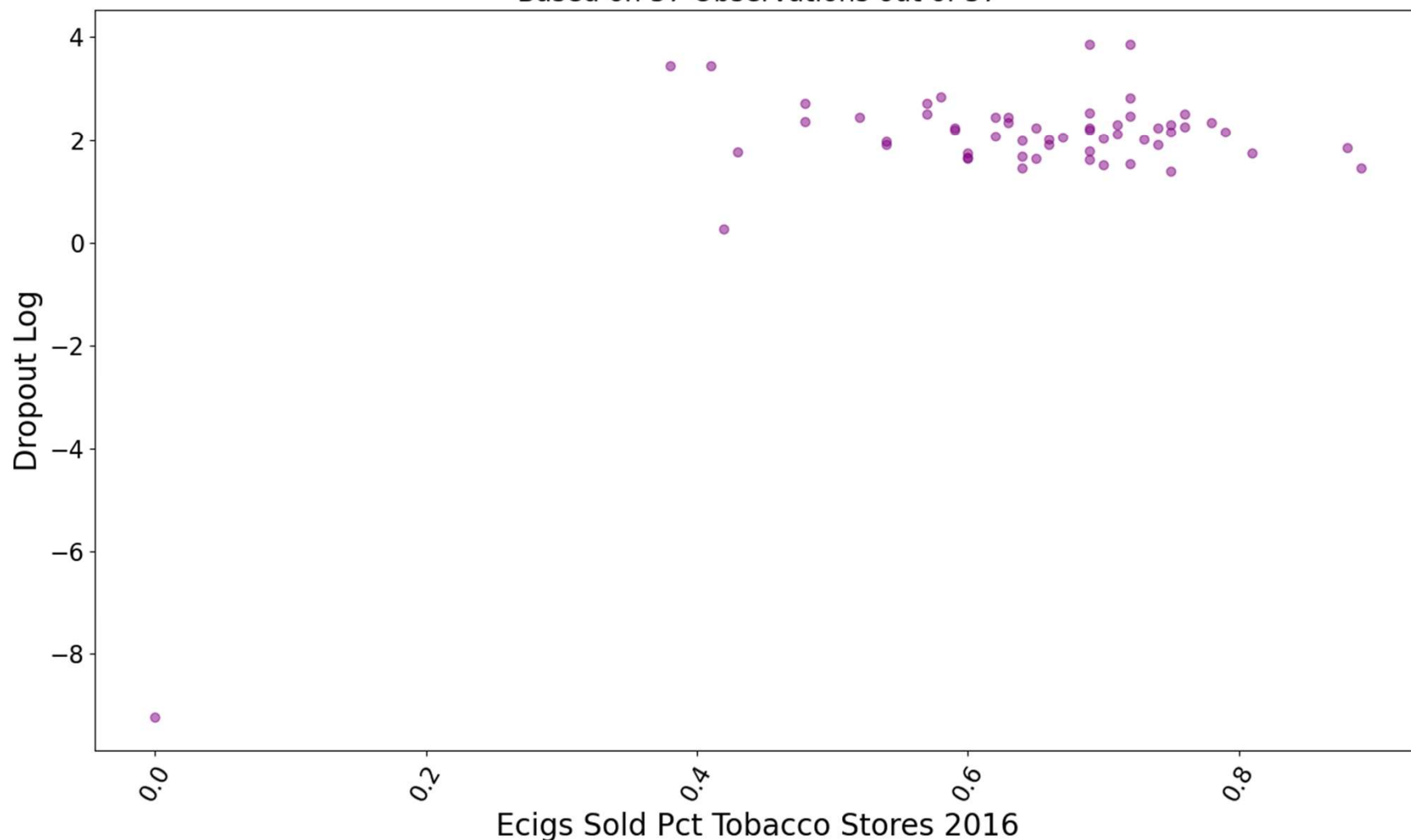
Based on 57 Observations out of 57



This is the distribution of ecigarette availability, or more specifically, the percent of tobacco stores that carried ecigarettes along with their other products.

## Relationship between Ecigs Sold Pct Tobacco Stores 2016 and Dropout Log

Based on 57 Observations out of 57



This is a scatterplot of how the availability of ecigarettes relates to the log-dropout rate. This plot shows some signs that the variables really do rise and fall together, but it also highlights the problem of outliers. Sierra County had a 0% dropout rate and a 0% ecigarette availability rate, meaning that it was the sole inhabitant of the bottom left corner, while all other observations fell more towards the top right. This outlier could be pulling the overall correlation towards a positive, rising line, even if the true relationship might be different and/or more complicated.